



Semester 1 Examinations 2019-2020

Exam Code(s)	1SD1, 1MDM1, 1IF1, 1MF1, 1MCR1, 1SD3
Exam(s)	Higher Diploma in Software Design and Development Higher Diploma in Software Design and Development, Industry Stream Masters in Software Design and Development Masters in Digital Media Masters Clinical Research
Module Code(s)	CT511
Module(s)	Database Development
Paper No.	
Repeat Paper	
External Examiner(s)	Professor Pier Luca Lanzi
Internal Examiner(s)	Professor Walter Gear Professor Michael Madden *Dr. Attracta Brennan

Instructions: Answer **questions 1 and 2** and 2 other questions
All questions carry equal marks

Duration	2 Hours
No. of Pages	7
Discipline(s)	Computer Science
Course Co-ordinator(s)	

Requirements:

MCQ		
Release to Library:	☒ X	Yes
Handout		No
Statistical/ Log Tables		☐
Cambridge Tables		
Graph Paper		
Log Graph Paper		

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QUESTION 1

Using the **EMPLOYEE** and **DEPARTMENT** tables,

EMPLOYEE

id	fname	lname	position	gender	salary	department
1	Ann	Harris	HR Admin	F	45000	D1
2	Sarah	Johnson	Accountant	F	43000	D2
3	Liam	Malarky	HR Manager	M	76000	D1
4	David	Oliver	Accountant	M	76000	D2
5	Jill	Leroy	Social Media	F	65000	D3
6	Jane	Simmons	PR	F	58000	D3
7	Aaron	Mills	Payroll	M	98000	D2

DEPARTMENT

departmentid	dname	tel
D1	HR	Ext 4532
D2	Finance	Ext 5411
D3	Marketing	Ext 3202
D4	Purchasing	Ext 4522
D5	Research	Ext 3097

Write the SQL for an MSSQL Server DB to :

1. Update the EMPLOYEE table for all those who work in Finance by changing the salary to 150% of the old salary.
2. Insert a new record into the EMPLOYEE table which contains the following information; Tom Thumb whose roles is in Sales and who is on a salary of 43000. Note that Sales are a part of the Marketing department.
3. Count the number of people in each department (note: watch your joins).

Department	Number of employees/department
D1	2
D2	3
D3	3
D4	0
D5	0

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4. Display those departments whose **TOTAL** salary spend is greater than 130,000.

Department	Total departmental salaries
D2	155850
D3	166000

5. Display the names as follows (note: the first letter in both the name and the surname are in lowercase while the remaining letters are in uppercase)

Full name
aNN hARRIS
sARAH jOHNSON
LIAM mALARKY
dAVID oLIVER
jILL lEROY
jANE sIMMONS
aARON mILLS
tOM tHUMB

6. Display the following based on department and gender (note: watch the ordering in the first two columns);

Department	Gender	Maximum salary	Minimum salary	Average salary
D1	F	45000	45000	45000
D1	M	76000	76000	76000
D2	F	64500	64500	64500
D2	M	51450	39900	45675
D3	F	65000	58000	61500
D3	M	43000	43000	43000

7. Display the following for all those whose first names start with J;

Records starting with J
Jill Leroy works in Marketing
Jane Simmons works in Marketing

8. Display the following;

Commentary
There are 2 staff working in D1 where the max. salary is 76000 and the average salary is 60500
There are 3 staff working in D2 where the max. salary is 147000 and the average salary is 108500
There are 3 staff working in D3 where the max. salary is 65000 and the average salary is 55333.3

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9. Display the following where you skip the first 2 records and then present only the next 3 records (note: watch the ordering);

NAME	SURNAME
David	Oliver
Aaron	Mills
Liam	Malarky

10. Display the first name, surname and salary of all staff whose salary is between the average and maximum of all salaries

(25 marks)

QUESTION 2

- a.) Draw the **EERD (Enhanced Entity Relationship Diagram)** for the following:

A community hospital has categorised person to be one of the following; patient, doctor and employee. Note: there may be other categories added at a later stage. An employee can also be a doctor. Each person is defined by their PPS, name, contact details and DOB. The doctor also has a specialisation-id.

A doctor **can be** responsible for many patients. It is mandatory that a patient is seen by a doctor. A patient is also defined by date admitted.

Patients must be either be outpatients or resident patients. In some cases they can be both. An outpatient can schedule at least one visit! A visit has to be scheduled by an outpatient. The visit is defined by visit id. A resident patient is provided with a bed, but a bed may/may not be allocated to a resident patient. Beds are defined by bed id and bed type.

An employee must be one of the following; a nurse, staff or technician. Note; these roles do not overlap.

Staff are employed in a work unit, whilst a work unit has multiple staff allocated to it. Work units are identified by id, location and contact details.

Nurses must be either Registered Nurse (RN) or Licensed Practise Nurse (LPN) only. Some nurses are assigned to multiple care centres, while others are not. A care centre will typically have multiple nurses assigned to it. A care centre is defined by id, address, specialisation.

A technician also works in a work unit. In general, work units have more than one technician allocated to them.

(17 marks)

- b.) Compare and contrast **file, record, field** and **primary key** outlining their importance (use examples).

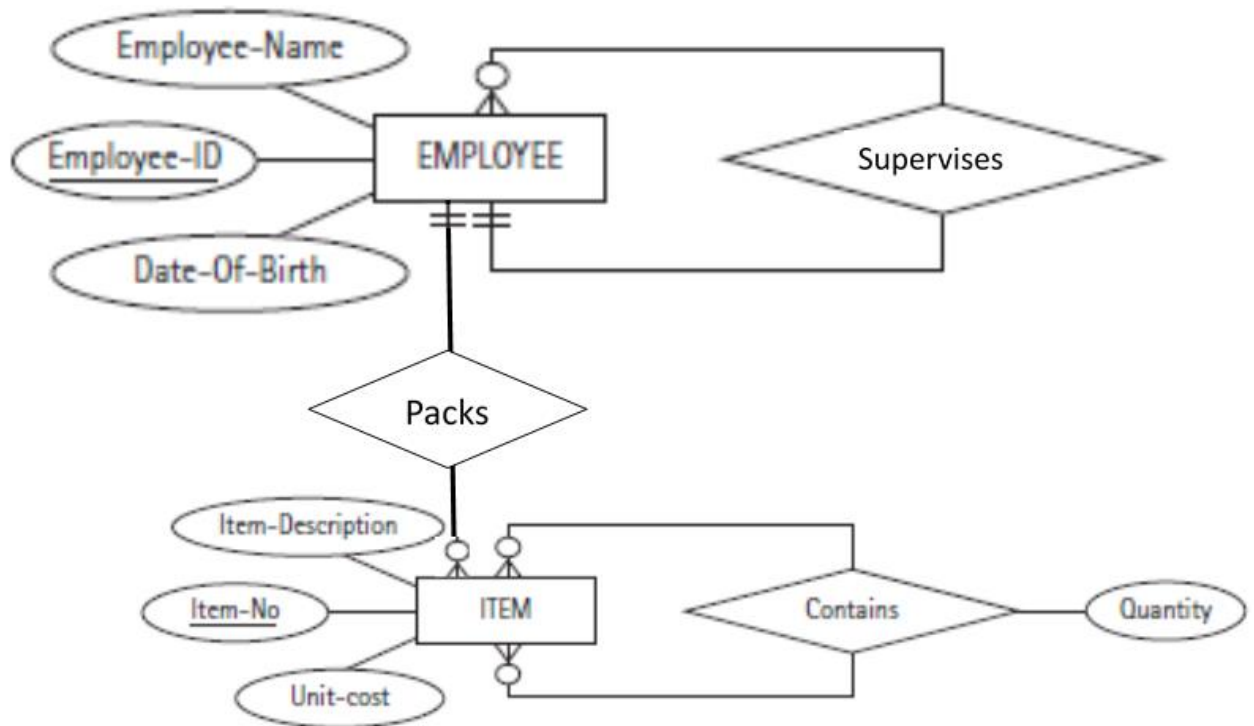
(8 marks)

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QUESTION 3

a.) The following is from a packaging plant. Map to relations using the short hand approach.

(10 marks)



b.) Discuss the 6 steps in the **System Development Life Cycle** for Database System design and development.

(9 marks)

c.) Compare and contrast the following; **composite**, **derived**, **stored** and **simple** attributes (use examples).

(6 marks)

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QUESTION 4

- a.) Explain why the following MOVIE table is not in 2NF. Bring it to 2NF by applying the 5 step approach. (15 marks)

MOVIE

CID	Name	Address	MovieID	Title	Category	Downloaded
1	Tom Jenkins	Galway	M1	Pirates of the Caribbean	Action	12/3/2020
2	Eli Linkin	Mayo	M1	Pirates of the Caribbean	Action	15/4/2020
3	Josh Mullins	Leitrim	M2	Forgetting Sarah	Romance	22/2/2020
3	Josh Mullins	Leitrim	M1	Pirates of the Caribbean	Action	1/1/2020
4	Ariel Mermaid	Galway Bay	M3	Undertales	Action	3/5/2020
1	Tom Jenkins	Galway	M5	Clash of the Titans	Action	3/5/2020

- b.) A car fleet loan company wants to create a simple EERD for its fleet of cars, which can comprise; either a Mercedes, BMW or Jaguar. Each car has mileage, license number, engine capacity, number of previous owners and cost associated with it. Note : The Jaguar cars have special vehicle operations which sub operations. These must be; luxury, performance and technology. These are discrete and non-overlapping. The Mercedes range has a specific safety feature, whilst the BMW has a proprietary entertainment system. Draw the EERD before generalisation and then after generalisation.

(7 marks)

- c) Explain the difference between the **Total** and the **Partial Specialisation** rules (use examples). (3 marks)

QUESTION 5

- a.) Answer the following;

What is the dependency called when the primary key of the relation TASK is the attribute TASK-No. Two other attributes of TASK are BUILDER-NAME and BUILDER-ADDRESS. The following functional dependency exists: BUILDER-NAME -> BUILDER-ADDRESS. Explain your reasoning.

Integrate the following two relations which were developed from two different ER diagrams:
CUST-VIEW1 (CUST-No, CUST-NAME, CUST-AGE)
CUST-VIEW2 (CUST-No, CUST-ADDRESS, CUST-PHONE)

Indicate the normal form. Decompose into 3NF if not in 3NF:

- MODULE (COURSE-No, SECTION-No)
- MODULE (COURSE-No, SECTION-No, LecturerID)
- MODULE (COURSE-No, SECTION-No, ROOM)
- MODULE (COURSE-No, SECTION-No, ROOM, CAPACITY)

(12 Marks)

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b.) Compare and contrast; **strong, weak** and **associative** entities (use examples).

(6 Marks)

c.) Compare and contrast **Conceptual/Logical** models with **Implementation** models (use examples).

(4 Marks)

d.) Compare and contrast **Disjointness Constraint** and **Completeness Constraint** (use examples).

(3 Marks)

QUESTION 6

a.) Define what is meant by **Denormalisation**. Then using examples, discuss the **3 Denormalisation Options**.

(10 Marks)

b.) Compare and contrast **Sequential, Hashed** and **Indexed** file organisation (use examples).

(9 Marks)

c.) Define what is meant by the following (use examples); **functional dependency diagram, determinant of, compound key**.

(6 Marks)

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